Landslide Susceptibility AI Assessment Report

Study Area: Central Italy Apennines (~4,100 sq km) | Total Slope Units: 7,360

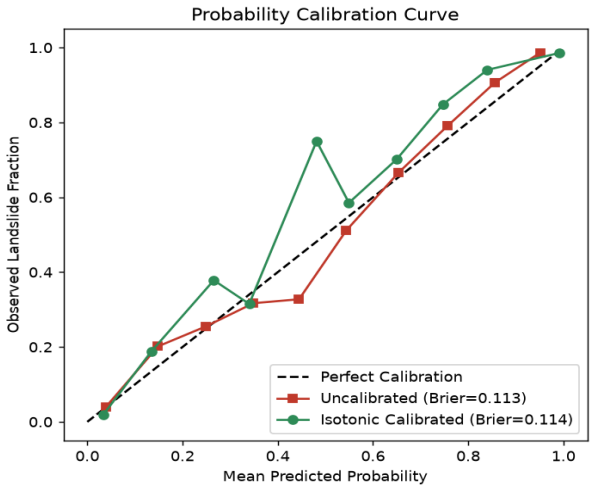
PROVENANCE NOTICE: *Synthetic Simulation (Central Italy parameters - non-real data). Simulation data for methodology demonstration. Not actual field measurements.*

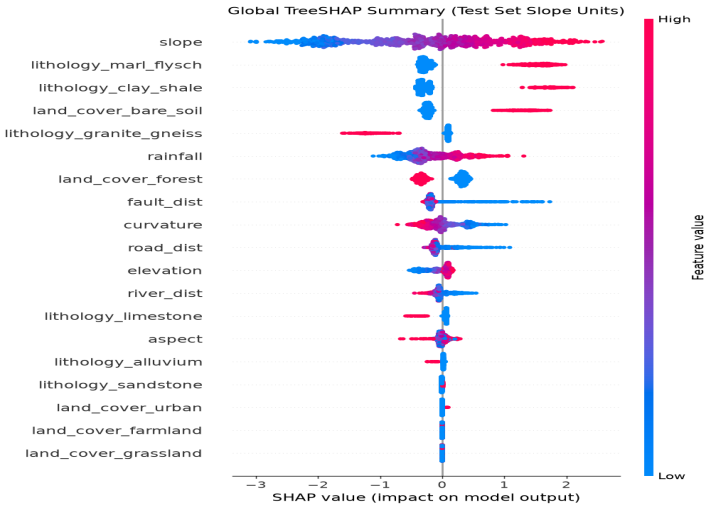
1. Model Comparison & Probability Calibration (Spatial Test Set)

Evaluated on held-out spatial blocks to prevent leakage. Brier score measures calibrated probability error:

Model	Accuracy	Precision	Recall	F1	AUC	Brier
Logistic Regression	0.841	0.851	0.808	0.829	0.918	0.114
Decision Tree	0.806	0.784	0.821	0.802	0.88	0.134
SVM	0.829	0.835	0.8	0.817	0.914	0.118
Random Forest	0.834	0.854	0.789	0.82	0.914	0.118
Proposed (XGBoost)	0.847	0.87	0.8	0.833	0.918	0.113
Proposed (XGBoost Calibrated)	0.848	0.881	0.789	0.832	0.917	0.114

2. Model Diagnostics (Calibration & SHAP Explainability)





3. Top 10 Riskiest Spatial Zones (Grid A1..E5)

zone	units	mean_risk	pct_high	main_rock	avg_slope	avg_rain
E3	289	0.776	79.9	clay_shale	26.7	1209.0
C2	305	0.758	74.1	clay_shale	29.0	1224.0
C3	284	0.718	70.8	marl_flysch	25.6	1267.0
A1	331	0.677	67.4	marl_flysch	24.2	1202.0
A3	278	0.616	55.0	sandstone	29.6	1327.0
D3	291	0.613	58.1	marl_flysch	27.3	894.0
A4	296	0.579	54.1	limestone	31.8	1281.0

C4	310	0.575	51.0	alluvium	30.3	966.0
B4	326	0.556	49.7	alluvium	28.5	1343.0
C1	302	0.542	50.7	clay_shale	21.4	1166.0

4. Top 10 Highest-Risk Slope Units

unit_id	zone	probability	risk_class	lithology	slope	rainfall	fault_dist
SU-0011	B2	1.0	Very High	marl_flysch	35.6	1200.0	6405.0
SU-7344	A1	1.0	Very High	marl_flysch	23.4	1392.0	9302.0
SU-7348	C2	1.0	Very High	clay_shale	32.2	1266.0	972.0
SU-0032	E1	1.0	Very High	sandstone	51.2	1215.0	21356.0
SU-7355	C4	1.0	Very High	marl_flysch	32.9	1009.0	2758.0
SU-0026	A4	1.0	Very High	alluvium	30.9	1383.0	718.0
SU-7358	A4	1.0	Very High	limestone	46.2	1380.0	2663.0
SU-1330	A5	1.0	Very High	limestone	54.7	1148.0	79.0
SU-1336	E3	1.0	Very High	clay_shale	40.5	1093.0	15398.0
SU-1344	C2	1.0	Very High	marl_flysch	26.6	1047.0	5666.0